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Enhanced Double Deep Q-Network (E-DDQN) with Asymmetric Reward Design Versus Standard Deep Q-Network for Safety-Prioritized Cybersickness Detection in Virtual Reality Under an Adversarial Threat Model: A Comparative Analysis

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ABSTRACT Immersive virtual reality (VR) is increasingly deployed in safety-critical settings such as clinical rehabilitation, training, and industrial simulation, where it exposes an underexamined perception-layer vulnerability: an adversary can manipulate rendering parameters (frame rate, field of view, latency, camera jitter) to induce cybersickness and cause direct physiological harm. Existing deep reinforcement learning (DRL) detectors optimize symmetric rewards that treat missed attacks and false alarms as equally costly, which is inappropriate when an undetected attack harms the user while a false alarm causes only a brief, recoverable interruption. This paper proposes the Enhanced Double Deep Q-Network (E-DDQN), a safety-prioritized dueling Double DQN that couples an asymmetric cost-sensitive reward, class-balanced experience replay, soft target updates, and validation-based recall-constrained model selection with post hoc SHAP explainability. On a held-out test set of 3,146 VR telemetry samples, E-DDQN attains 96.88% accuracy, 96.43% precision, 99.53% recall, 88.95% specificity, 94.24% balanced accuracy, an F1 score of 97.96%, and a Matthews correlation coefficient (MCC) of 0.916, missing only 11 of 2,359 attacks. Against a standard DQN baseline (94.53% accuracy, 94.57% recall, 0.861 MCC, 128 missed attacks), E-DDQN improves accuracy, recall, F1, and MCC simultaneously and reduces missed attacks by 91%, intentionally accepting a modest rise in false positives to achieve near-complete detection of the positive (cybersickness-event) class. Performance is stable across five random seeds and the gain over the baseline is statistically significant (McNemar, $p < 0.001$); a leave-one-out ablation attributes the safety gain to the asymmetric reward and the accuracy gain to class-balanced replay and the soft target update. A train-to-test accuracy gap of about 1.2 percentage points confirms strong generalization. Because the binary Attack target is a proxy derived from cybersickness-severity grades, these findings establish safety-prioritized detection under an adversarial threat framing rather than demonstrated robustness to explicit input perturbations, which remains future work.

INDEX TERMS Deep reinforcement learning, double deep Q-network, adversarial threat modelling, virtual reality, cybersickness, asymmetric reward, cost-sensitive learning, explainable AI, SHAP, safety-critical systems.

I. INTRODUCTION

Immersive virtual reality (VR) has moved from entertainment into operational, safety-critical settings such as clinical

rehabilitation, industrial and defence training, and human-performance assessment, where the headset actively shapes perception, balance, and spatial orientation rather than merely presenting information on a screen. This is precisely what makes VR vulnerable in a way conventional software is not: an adversary who manipulates rendering parameters such as frame rate, field of view, motion-to-photon latency, or camera jitter can deliberately induce cybersickness (nausea, disorientation, oculomotor strain, and, in severe cases, postural instability) and thereby cause direct physiological harm [1], [2]. Long studied as an ergonomic side effect of sensory conflict, cybersickness thus becomes a perception-layer attack surface. We use the term *adversarial cybersickness* for the deliberate induction, amplification, or exploitation of these symptoms to degrade user performance, impair situational awareness, or evade automated safety monitoring [1], [3].

AI-based detection has advanced from questionnaire instruments and classical classifiers (support vector machines, random forests) to deep temporal models (CNN-LSTM hybrids and attention- or transformer-based fusion) that better capture the multimodal, time-varying structure of VR telemetry and improve benign-condition accuracy [4], [5], [6], [7], [8], [9]. Three limitations nonetheless persist. First, most detectors are static classifiers trained offline and applied without policy adaptation, so they cannot track adversarial patterns that evolve within a session [10], [11]. Second, they are evaluated under non-adversarial assumptions, so high reported accuracy does not imply robustness to deliberate perturbations in motion, field of view, or latency [2]. Third, explainability, where present, often adds inference latency incompatible with the roughly 20 ms VR rendering budget [2], [3].

These gaps are sharper for reinforcement-learning detectors. Deep reinforcement learning (DRL) is attractive because it models detection as sequential decision-making, updating a policy through interaction rather than fitting a fixed distribution, a conceptual advantage when attacks are dynamic. Yet the standard Deep Q-Network (DQN) is ill-suited to safety-critical detection in its basic form: it suffers overestimation bias because action selection and value estimation share one network [12], and, more importantly, it optimizes a symmetric reward that prices a missed attack and a false alarm equally. That assumption is invalid here, where a false negative leaves an adversarial attack active and the user exposed to harm, whereas a false positive causes only a brief, recoverable interruption [2], [13]. No prior framework unifies adaptive policy learning, safety-aware reward design, and post hoc interpretability for adversarial perceptual threats in VR.

To close this gap we propose the Enhanced Double Deep Q-Network (E-DDQN), a safety-prioritized dueling Double DQN that (i) decouples action selection from value estimation and state value from action advantage to curb overestimation; (ii) encodes an asymmetric cost-sensitive reward that penalises missed attacks far more than false alarms; (iii) stabilises learning on imbalanced data through class-balanced experience replay and soft (Polyak) target updates; (iv) selects the deployed model by a validation-based, recall-constrained criterion; and (v) attaches a post hoc SHAP module that makes every decision auditable at no inference-time cost. On the Gaze-Locomotion Cybersickness Dataset (GLCD; 20,972 samples, 41 features), E-DDQN reaches 96.88% accuracy and 99.53% recall, missing only 11 of 2,359 attacks versus 128 for a standard DQN baseline (a 91.4% reduction) while improving accuracy, F1, and MCC simultaneously and conceding only a small precision and specificity margin.

This work makes four contributions. (1) It frames cybersickness detection as a safety-prioritized decision problem motivated by adversarial perceptual threats in VR, rather than a purely ergonomic classification task. (2) It formulates detection as a sequential decision process and introduces a safety-prioritized, cost-sensitive reward whose asymmetric penalties make missed attacks far more costly than false alarms. (3) It shows that a dueling Double DQN trained with this reward, class-balanced replay, soft target updates, and validation-based recall-constrained selection improves accuracy, recall, F1, and MCC over a standard DQN baseline while cutting missed attacks by 91%, with a leave-one-out ablation isolating each component's contribution. (4) It integrates post hoc SHAP explainability for per-decision auditing without inference-time overhead. The remainder of the paper is organised as follows: Section II reviews related work; Section III details the E-DDQN methodology; Section IV the experimental setup; Section V reports and discusses results; and Section VI concludes.

II. RELATED WORK

Research relevant to adversarial cybersickness detection spans four streams: AI-based cybersickness detection, adversarial VR threat modelling, explainable AI for safety-critical systems, and deep reinforcement learning for sequential detection. Each has advanced independently; none unifies all four into an adaptive, explainable, and safety-prioritized detector. We synthesise each stream and the cumulative gap that motivates E-DDQN.

A. AI-BASED CYBERSICKNESS DETECTION

The Simulator Sickness Questionnaire [14] remains the standard psychometric instrument but measures discomfort post hoc and cannot capture symptom onset during active immersion [15], motivating sensor-driven AI. Early classical classifiers on handcrafted physiological features (SVMs on galvanic skin response and heart rate [4] and random forests on electrodermal, cardiac, and inertial signals [5]) reached roughly 85% but depend on expert feature engineering and ignore temporal evolution. Deep temporal models followed: BiLSTM with GAN-based augmentation [16], a CNN-ECA-LSTM reaching 92% on EEG [6], convolutional networks on eye-tracking signals [17], and attention- and transformer-based fusion of EEG, IMU, and eye-tracking signals that cut latency and improved out-of-session generalization [7], [8], [9]. Across all of these, however, no study evaluates

performance under deliberate adversarial perturbation, and only one couples detection with explainability Table 1.

B. ADVERSARIAL VR THREATS

VR mediates spatial perception and vestibular alignment, exposing an attack surface distinct from conventional systems [18], [19]. Tseng et al. [1] showed that subtle changes to optical flow, spatial frequency, and binocular disparity induce nausea and disorientation without tripping conventional intrusion detection, and Kundu et al. [2] formalised an adversarial-cybersickness threat model spanning rendering-level distortion, locomotion manipulation, and sensor-pipeline attacks. Operational VR-security taxonomies enumerate cybersickness induction as a verifiable threat [20], [21] but propose no real-time automated detector, and network-layer RL defences [22] do not address physiological, perception-layer attacks. Cybersickness and VR-security research thus remain largely separate, leaving adversarial cybersickness detection unaddressed [3], [23].

C. EXPLAINABLE AI FOR SAFETY-CRITICAL DETECTION

Black-box accuracy is insufficient in regulated VR deployment, where decisions must be auditable [3]. SHAP [24] dominates attribution through theoretically grounded, per-instance feature importance, complemented by LIME [25]. TruVR [26] and LiteVR [27] integrated SHAP into VR cybersickness monitoring (91.3% and 89.7%) but were never tested against deliberate perturbations; the most directly related study [2] couples cybersickness-attack detection with explainability but reports that SHAP-based explanation incurs substantial per-inference overhead, exceeding the $\sim$20 ms VR rendering budget [2]. Cross-domain work confirms XAI reliably flags manipulated inputs [28]. No prior framework simultaneously achieves adversarial robustness, real-time inference, and explainability Table 2.

D. DEEP REINFORCEMENT LEARNING FOR SEQUENTIAL DETECTION

Unlike static classifiers, DRL agents learn a detection policy through sequential interaction [12], [29], matching streaming VR telemetry in which attacks may sustain or escalate across a session. The original DQN [12] suffers from overestimation bias because one network both selects and evaluates the bootstrap action; Double DQN [13] corrects this by decoupling selection and evaluation across two networks, the foundational component of E-DDQN. DRL detectors succeed in network intrusion and anomaly prioritisation [22], but all use symmetric rewards that weight false negatives and false positives equally. Safe-RL research shows that asymmetric penalties yield more risk-conservative policies [29], yet asymmetric reward design has not been applied to DRL-based VR cybersickness detection Table 3.

E. IDENTIFIED RESEARCH GAP

Synthesising the four streams exposes five interlocking gaps: the near-absence of adversarial-robustness evaluation in cybersickness detection; the dominance of static classifiers that cannot adapt their policy; the symmetric reward of all prior DRL security frameworks; the unresolved conflict between XAI transparency and real-time inference; and the lack of any single framework addressing all four Table 4. E-DDQN targets every gap with a sequential DRL policy, an asymmetric safety-aware reward, class-balanced replay and soft updates for stable imbalanced learning, recall-constrained model selection, and post hoc SHAP at no inference cost, a combination absent from the prior cybersickness or VR-security literature [2], [3], [13].

III. PROPOSED METHODOLOGY

E-DDQN casts adversarial cybersickness detection as a sequential decision problem solved by a dueling Double DQN under a safety-prioritized objective. Fig. 1 traces the end-to-end training and detection procedure: VR telemetry is preprocessed into a 41-dimensional state vector, the agent emits a binary Normal/Attack action, and the same trained network feeds a parallel post hoc SHAP module that explains each decision. Five components distinguish E-DDQN from a standard DQN: a dueling Double-DQN value head, an asymmetric cost-sensitive reward, class-balanced experience replay, soft target updates, and validation-based recall-constrained selection, each detailed below and isolated by ablation in Section V-F.

A. PROBLEM FORMULATION AS A MARKOV DECISION PROCESS

Detection is modelled as a Markov decision process $(\mathcal{S}, \mathcal{A}, \mathcal{R}, \gamma)$ following the value-based formalism of Sutton and Barto [29]. A state $s_t \in \mathcal{S}$ is the VR-telemetry feature vector at decision step t,

$$s_t = [f_1, f_2, \ldots, f_{41}]^T \in \mathbb{R}^{41} \tag{1}$$

The action space is binary, $\mathcal{A} = \{0, 1\}$: action 1 raises an Attack alert and action 0 declares the sample Normal. Collapsing the original four ordinal discomfort grades into this binary space reduces the policy to the operationally relevant decision of whether to raise an alert. The agent maximizes the expected discounted return

$$G_t = \sum_{k=0}^{\infty} \gamma^k r_{t+k}, \quad \gamma \in [0, 1) \tag{2}$$

Acting greedily on a learned action-value function $Q(s, a)$, the optimal value obeys the Bellman optimality equation

$$Q^*(s, a) = \mathbb{E}\big[r + \gamma \max_{a'} Q^*(s', a') \mid s, a\big] \tag{3}$$

During training the agent follows an ε-greedy policy that balances exploration and exploitation,

$$a_t = \arg\max_a Q(s_t, a; \theta) \text{ w.p. } 1-\varepsilon, \text{ else a random action in } \mathcal{A} \tag{4}$$

with ε annealed from 1.0 to 0.01. Because the policy is learned by reward-driven interaction rather than fixed offline,

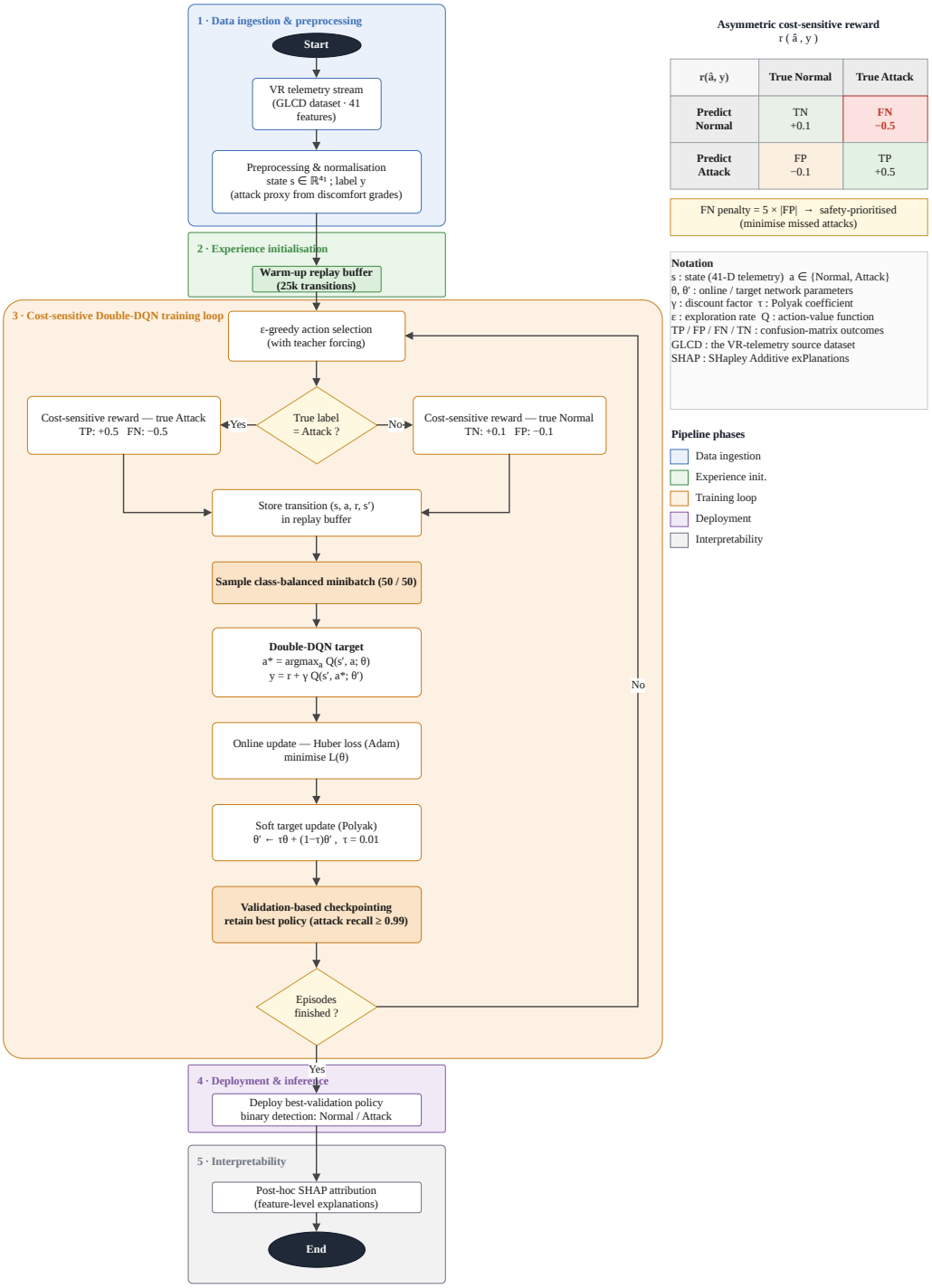


FIGURE 1. End-to-end E-DDQN training and detection procedure. VR telemetry is preprocessed into the 41-D state vector; under an ε-greedy policy with teacher forcing the agent receives the asymmetric cost-sensitive reward (TP $+0.5$ / FN -0.5 / TN $+0.1$ / FP -0.1; inset matrix, with the FN penalty five times the FP penalty), stores the transition, and updates the dueling Double-DQN online network from class-balanced 50/50 minibatches followed by a soft Polyak target update ($\tau = 0.01$). Validation-based, recall-constrained selection (recall ≥ 0.99) checkpoints the best model; the deployed model performs Normal/Attack detection followed by post hoc SHAP attribution. The binary Attack label is a proxy derived from the GLCD discomfort grades.

TABLE 1. Comparison of AI-based cybersickness detection methods (2016–2025). Adv. Test = adversarial robustness evaluation; XAI = explainability integration. For the proposed method, "Proxy" denotes evaluation on proxy attack labels under an adversarial threat model rather than explicit perturbation testing.

Study	Sensing Modality	Model Type	Accuracy	Adv. Test	XAI	Key Limitation
Dennison et al. [4]	EDA, HR	SVM	∼85%	✗	✗	No temporal modelling
Islam et al. [5]	EDA, HRV, IMU	Random Forest	∼85%	✗	✗	Static feature engineering
Qu et al. [17]	Eye tracking	CNN	∼88%	✗	✗	Single modality, no XAI
Yalçin & Halbig [16]	EDA, HR, HRV	BiLSTM + GAN aug.	Real-time	✗	✗	Requires wearable sensors
Liu et al. [6]	EEG (8-channel)	CNN-ECA-LSTM	92%	✗	✗	Clinical EEG only
Jeong et al. [7]	EEG, IMU, Eye	MAC Attn-CNN-LSTM	∼88%	✗	Partial	High compute, 3 sensors needed
Jeong & Han [8]	Multimodal fusion	Transformer	SOTA	✗	✗	No adversarial framing
Ma et al. [9]	Head motion, Eye	Hierarchical Attn	SOTA	✗	✗	Benign conditions only
Kundu et al. [2]	IMU, Eye tracking	CNN-LSTM + SHAP	93.5%	✓	✓	SHAP adds ∼35 ms latency
E-DDQN (Proposed)	VR telemetry (41 features)	DDQN + SHAP	**96.88%**	**Proxy**	✓	Higher FP rate

TABLE 2. Explainable-AI methods applied to VR security and cybersickness detection. Adv. Eval. = adversarial evaluation; opt. = optimized.

Study / Framework	XAI Method	Adv. Evaluation	Accuracy	Latency	Key Limitation
Kundu et al. [26], TruVR	SHAP	✗ None	91.3%	Not reported	No adversarial robustness test
Kundu et al. [27], LiteVR	SHAP + Pruning	✗ None	89.7%	Edge-opt.	Benign data only
Kundu et al. [2]	SHAP + FGSM/PGD	✓ FGSM, PGD	93.5%	∼35 ms	Latency exceeds VR threshold
Karapiperis et al. [28]	SHAP + LIME	✓ Cyber domain	98.5%	Not reported	Not validated in VR context
Jeong et al. [7], MAC	Attention maps	✗ None	∼88%	∼100 ms	No adversarial framing
E-DDQN (Proposed)	SHAP DeepExplainer	Proxy labels	**96.88%**	Not profiled	Higher FP rate

TABLE 3. Deep reinforcement learning architectures for security and adversarial detection. Overest. Fix = overestimation-bias correction; Asym. Reward = asymmetric cost-sensitive reward; VR App. = validated in VR.

Framework	Domain	Reward Type	Overest. Fix	XAI	VR App.	Key Limitation
Standard DQN [12]	Game / General	Symmetric	✗	✗	✗	Overestimation bias
Double DQN [13]	Game / General	Symmetric	✓	✗	✗	No safety-aware reward
Federated DDQN [22]	Network IDS	Symmetric	✓	✗	✗	No VR or XAI support
RL Anomaly Prioritisation [22]	DDoS Detection	Symmetric	✗	✗	✗	Network layer only
Safe RL [29]	General Safety	Asymmetric	Partial	✗	✗	Not applied to VR
E-DDQN (Proposed)	**VR (safety-critical)**	**Asymmetric 5:1**	✓	✓	✓	Higher FP rate

TABLE 4. Identified research gaps, supporting evidence, operational consequence, and corresponding E-DDQN design requirement.

Gap Identified	Evidence from Literature	Operational Consequence	E-DDQN Design Requirement
Absence of adversarial robustness evaluation	Most reviewed studies; only Kundu et al. [2] provide adversarial testing	Reported accuracy not guaranteed under deliberate sensor attack	Safety-prioritized DRL detection under an adversarial threat model; explicit perturbation-based tests (FGSM/PGD) are future work
Static classifiers; no adaptive detection policy	Caserman et al. [10]; Yang et al. [11]	Models degrade under evolving or novel adversarial patterns	DRL sequential policy with online reward-driven update
Symmetric reward ignores asymmetric error costs	Mnih et al. [12]; van Hasselt et al. [13]	FN and FP treated equally despite unequal physiological harm	Asymmetric cost-sensitive reward: FN penalised 5× FP
XAI latency exceeds real-time VR constraint	Kundu et al. [2]; Nair et al. [3]	Explainability and operational speed are mutually exclusive	Post-training SHAP integration with no inference-time overhead
No unified adversarial-robust, XAI-enabled DRL framework for VR	All reviewed adversarial and XAI studies	Fragmented protection with no single deployable solution	E-DDQN: SHAP + asymmetric reward + DDQN architecture for VR

the detector can in principle keep adapting to shifting adversarial patterns, addressing the static-classifier limitation of Section II.

B. DATASET AND PREPROCESSING

Experiments use the Gaze-Locomotion Cybersickness Dataset (GLCD): 20,972 VR-telemetry samples described by 41 physiological and motion-related features, supplied pre-split into disjoint training (14,680), validation (3,146),

and test (3,146) partitions so that the test partition is never observed in training and no information leaks across the split. The original labels are four ordinal discomfort grades (0–3), balanced in training (3,670 per grade). For binary detection, any non-zero grade maps to Attack and grade 0 to Normal, giving 3,670 Normal / 11,010 Attack in training and 787 / 2,359 at test Fig. 3, an approximately 3:1 imbalance that motivates the class-balanced replay of Section III-F. Features are standardised with robust quartile-based scaling to limit the heavy-tailed motion outliers common in head-mounted-display telemetry [30]; tensors are stored in single precision and kept separate from the replay buffer to preserve the feature dimensionality the SHAP module requires Fig. 4. In this study the binary Attack target is a proxy derived from the discomfort grades; the threat model of Fig. 2 (latency/frame-pacing, motion distortion, and field-of-view manipulation) frames the perception-layer attack surface, while explicit perturbation generation (FGSM/PGD) is left to future work.

C. DUELING VALUE-FUNCTION APPROXIMATOR

Both the evaluation network (parameters θ) and the target network (parameters θ^-) share a dueling architecture [31] built on an hourglass multilayer-perceptron trunk of widths $41 \to 256 \to 512 \to 256$. Each hidden layer applies batch normalisation [32] to stabilise activations under the non-stationary RL updates, a rectified-linear nonlinearity, and dropout of 0.2 [33] to limit overfitting to the majority class; weights use Xavier initialisation [34]. The trunk splits into a state-value stream $V(s)$ and an advantage stream $A(s,a)$ that recombine as

$$Q(s,a) = V(s) + \left[A(s,a) - \frac{1}{|\mathcal{A}|}\sum_{a'} A(s,a')\right] \tag{5}$$

Subtracting the mean advantage makes the decomposition identifiable and ties V to the state value; decoupling value from advantage yields more stable estimates on states where the action choice is nearly inconsequential (most benign samples), improving specificity. The $\arg\max$ over the two Q-values is the detection decision (Fig. 5). The trunk is deliberately compact to meet the real time inference budget of the VR rendering pipeline.

D. DOUBLE DQN LEARNING WITH OVERESTIMATION CORRECTION

Standard DQN [12] uses one network to both select and evaluate the bootstrap action, coupling selection noise to value estimation and inflating Q-values. Double DQN [13] decouples the roles: the online network selects the greedy next action and the target network evaluates it, giving the temporal-difference target

$$y = r + \gamma \cdot Q\big(s', \arg\max_{a'} Q(s',a';\theta);\theta^-\big) \tag{6}$$

The evaluation network minimises the Huber (smooth-L1) loss between predicted value and target,

$$\mathcal{L}(\theta) = \mathbb{E}\big[\mathcal{L}_\delta\big(y - Q(s,a;\theta)\big)\big],$$
$$\mathcal{L}_\delta(e) = \begin{cases} \frac{1}{2}e^2, & \text{if } |e| \le \delta, \\ \delta\left(|e| - \frac{1}{2}\delta\right), & \text{otherwise.} \end{cases} \tag{7}$$

which is quadratic for small errors and linear for large ones, damping the destabilising effect of the large temporal-difference errors produced by random transitions early in training [35]. The target network tracks the online network by soft (Polyak) averaging after every learning step,

$$\theta^- \leftarrow (1-\tau)\cdot\theta^- + \tau\cdot\theta, \quad \tau = 0.01 \tag{8}$$

which replaces the periodic value jumps of hard synchronisation with a smoothly moving bootstrap target [36]. Optimisation uses Adam [37] (learning rate 10^{-3}, weight decay 10^{-4}) with a cosine-annealed schedule toward 10^{-5} [38] and gradient-norm clipping at 1.0. Transitions are stored in a circular replay buffer of 50,000 and sampled to decorrelate consecutive updates [39].

E. ASYMMETRIC COST-SENSITIVE REWARD

The reward function is the principal contribution. Symmetric value-based detectors reward a correct decision $+1$ and an error -1, implicitly pricing a missed attack and a false alarm equally. In safety-critical VR this is invalid: a false negative leaves an adversarial attack active and the user exposed to physiological harm, whereas a false positive triggers only a brief, recoverable interruption [1], [2]. E-DDQN therefore assigns outcome-dependent rewards following the cost-sensitive-learning principle [40]:

$$r = +0.50\ (\text{TP}),\ +0.10\ (\text{TN}),\ -0.50\ (\text{FN}),\ -0.10\ (\text{FP}) \tag{9}$$

This sets a $5:1$ penalty ratio between false negatives and false positives, so the return-maximising policy aggressively suppresses missed attacks even at the cost of additional false alarms; rewarding correct detections five times more strongly than correct rejections further counters the residual majority-class pressure (Fig. 6). The reward is the only change to the agent objective relative to the baseline, so any shift in the recall-precision balance is attributable directly to the cost structure. The penalty ratio is a tunable deployment lever (Section V-G), not a universal constant.

F. CLASS-BALANCED EXPERIENCE REPLAY

Uniform replay sampling would reproduce the approximately 3:1 Attack:Normal imbalance in every minibatch, biasing the agent toward over-alerting and depressing specificity. E-DDQN instead draws each minibatch of size B with a fixed class composition,

$$n_{\text{Normal}} = \rho B, \quad n_{\text{Attack}} = (1-\rho)B, \quad \rho = 0.5 \tag{10}$$

sampling the two outcome classes independently from the buffer (with replacement when a class is scarce). Equalising



FIGURE 2. Adversarial threat model: three conceptual attack classes (latency/frame-pacing, motion distortion, and field-of-view manipulation) defining the perception-layer attack surface that motivates detection.

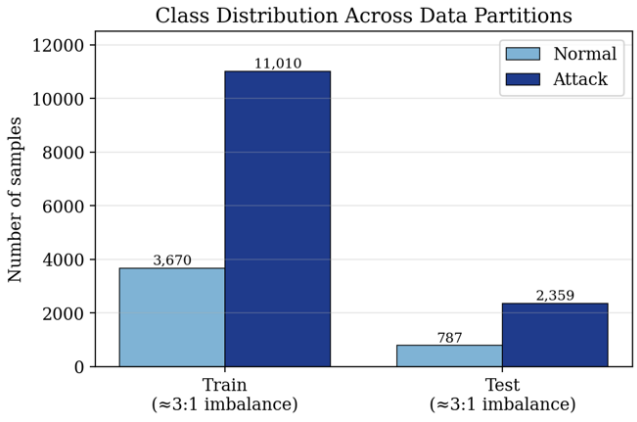


FIGURE 3. Class distribution across the training and test partitions, showing the approximately 3:1 Attack-to-Normal imbalance.

the classes lets the agent learn the Normal decision boundary as thoroughly as the Attack one, reducing false positives at a fixed recall; the ablation in Section V-F confirms that class-balanced replay is the principal driver of aggregate accuracy and specificity.

G. TEACHER-FORCED EXPLORATION

Early ε-greedy exploration injects many random actions into the buffer; stored naively, these carry no supervisory signal. E-DDQN overwrites the stored action of an exploratory transition with the ground-truth label before buffering it, while greedy (network-selected) actions are stored unchanged. The reward is always computed from the action actually taken, so the safety incentives of (9) are preserved. Adapted from teacher forcing in sequence learning [41], this keeps the exploration phase informative and accelerates early value-function convergence.

H. WARM-UP REPLAY-MEMORY INITIALISATION

Before any gradient step the replay buffer is pre-filled with random transitions until half full (25,000), using the asymmetric reward of (9). Without warm-up the first updates would draw from a small, highly correlated sample and produce unstable gradients; warming the buffer guarantees a diverse, decorrelated population from the first update, a standard stabilisation device for off-policy value learning [39].

I. TRAINING PIPELINE AND MODEL SELECTION

For efficiency the agent processes telemetry in vectorised chunks of 256 samples (performing ε-greedy selection, reward assignment, teacher forcing, and buffer insertion for an entire chunk at once) and applies a batch-512 update every 50 chunk steps plus five consolidation updates per episode, over 500 episodes with ε decayed at 0.97 per episode. This reduces forward passes per episode by roughly two orders of magnitude relative to single-sample processing without

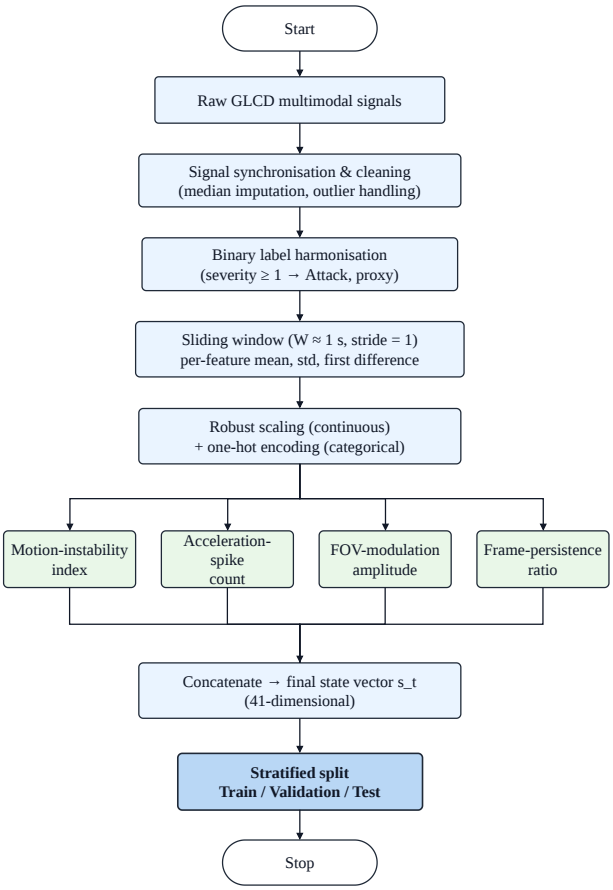


FIGURE 4. Preprocessing and feature-engineering pipeline transforming raw GLCD signals into the 41-dimensional state vector through cleaning, temporal windowing, scaling, and encoding.

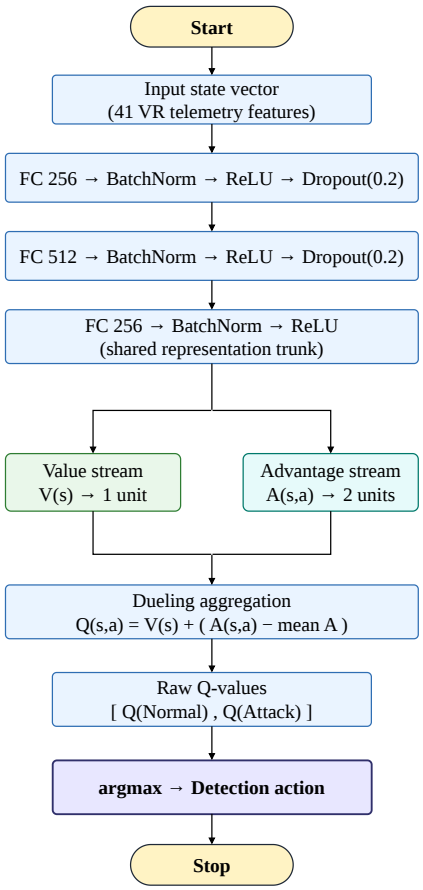


FIGURE 5. Dueling Double-DQN network: a shared 256-512-256 trunk feeds separate value and advantage streams that recombine through the dueling aggregation into two raw Q-values.

altering the learning dynamics, making the repeated experimentation required for reward design and ablation practical. The deployed model is chosen by a validation-based, recall-constrained rule (Fig. 7): a checkpoint is retained only when its validation recall meets a 0.99 floor and its validation F1 improves, and the held-out test set is evaluated exactly once, on the final selected model, eliminating test-set leakage. Algorithm 1 summarises the procedure.

J. POST-HOC SHAP EXPLAINABILITY

To make the policy auditable, a SHAP DeepExplainer [24] is applied after training, using the first 100 training samples as the background reference and attributing the Attack-class output across test samples. Because the attribution is computed offline, after the detection decision, it adds no latency to the inference path and so does not violate the VR real time constraint [2]. The mean absolute Shapley value per feature gives a global ranking of the telemetry channels that drive the adversarial decision, and the per-sample values support case-level auditing, both absent from the standard DQN baseline.

K. STANDARD DQN BASELINE

The baseline is a standard DQN [12] aligned with E-DDQN in architecture ($41 \rightarrow 256 \rightarrow 512 \rightarrow 256 \rightarrow 2$), replay capacity, target-update frequency, discount factor, and Adam optimizer, but differing from it in several respects at once: it uses a single network for selection and evaluation, a symmetric $+1/-1$ reward, uniform replay, hard target updates every 10 steps, a step learning-rate schedule, and single-sample processing, with last-episode selection. The two models therefore differ in several factors simultaneously (Table 5), so the head-to-head result in Section V-C compares the two complete configurations rather than isolating a single variable; the leave-one-out ablation of Section V-F (Table 11) supplies the per-component attribution.

IV. EXPERIMENTAL SETUP

A. IMPLEMENTATION AND HYPERPARAMETERS

Both models are implemented in PyTorch and trained on identical partitions with a fixed seed (42) governing the PyTorch, NumPy, and Python generators and deterministic CUDA settings, so the reported results are reproducible. Table 5 lists

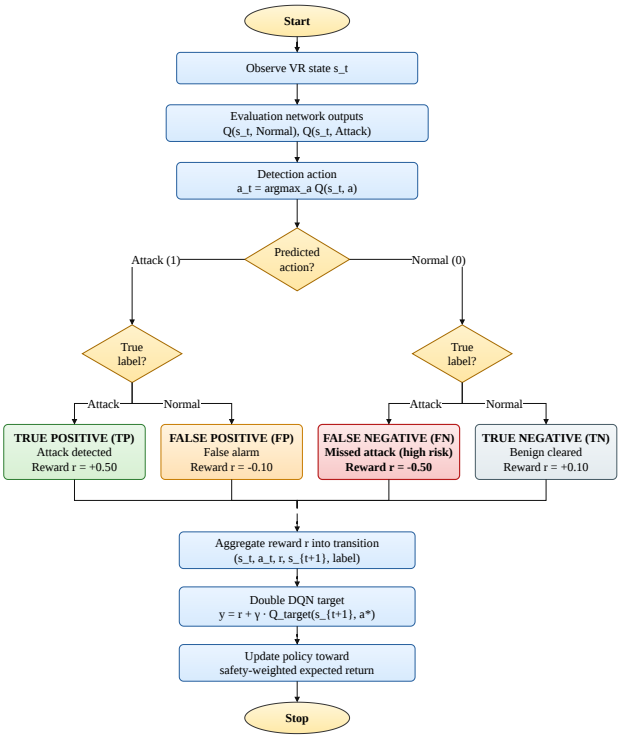


FIGURE 6. Detection decision and asymmetric reward mechanism: the arg max over the two Q-values yields the action, comparison with the true label produces the four outcomes and their rewards, and these feed the Double DQN update.

Algorithm 1 E-DDQN training with validation-based selection.

Initialise eval net θ, target net $\theta^- \leftarrow \theta$, replay buffer $\mathcal{D}$ (cap 50k)
Warm-up: fill $\mathcal{D}$ to 25k random transitions, reward by Eq. (9)
for episode $= 1$ to 500 **do**
  shuffle training set; $\varepsilon \leftarrow \max(0.01, 0.97 \cdot \varepsilon)$
  for each chunk of 256 samples **do**
    select actions by ε-greedy Eq. (4); reward by Eq. (9)
    teacher-force exploratory actions; store in $\mathcal{D}$
    every 50 chunks: sample class-balanced batch Eq. (10); update θ by Eqs. 6–7; soft-update θ^- by Eq. (8)
    5 consolidation updates
  end for
  every 10 episodes: evaluate on validation; keep θ if recall ≥ 0.99 and F1 improved
end for
load best-validation θ; evaluate once on test; run SHAP

the hyperparameters side by side: the upper rows list the settings shared by both models, while the lower rows mark the components that distinguish E-DDQN from the baseline.

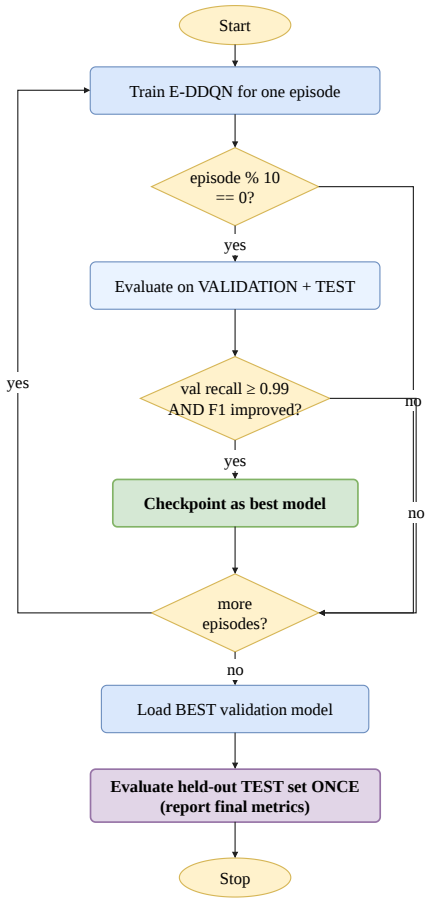


FIGURE 7. Validation-based, recall-constrained model-selection workflow. Periodic validation governs checkpointing under a 0.99 recall floor; the held-out test set is never used for selection and is reported only for the final model.

B. EVALUATION METRICS

Detection quality is reported with accuracy, precision, recall, F1, and the Matthews correlation coefficient (MCC), the last robust to class imbalance [42], derived from the confusion-matrix counts of true positives (TP), true negatives (TN), false positives (FP), and false negatives (FN) on the held-out test set, with Attack as the positive class:

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \tag{11}$$

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}, \quad \text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{12}$$

$$\text{F1} = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \tag{13}$$

Consistent with the safety-critical objective, recall (equivalently one minus the false-negative rate) is the primary metric, because every false negative is an adversarial attack that reaches the user; precision and accuracy are secondary metrics quantifying the false-alarm cost incurred for higher recall. The raw count of missed attacks (FN) is reported throughout,

TABLE 5. Hyperparameters and components of E-DDQN and the standard DQN baseline.

Hyperparameter / Component	E-DDQN (proposed)	Standard DQN (baseline)
Network architecture	41-256-512-256-2 MLP	41-256-512-256-2 MLP
Replay buffer capacity	50,000	50,000
Target update	soft (Polyak, τ=0.01)	hard, every 10 steps
Discount factor (γ)	0.99	0.99
Optimizer	Adam (lr 0.001, wd 1e-4)	Adam (lr 0.001)
Exploration (ε)	1.0 $\rightarrow$ 0.01, decay 0.97	1.0 $\rightarrow$ 0.01, decay 0.97
Episodes	500	150
Q-update rule	Dueling Double DQN	Standard DQN
Reward function	Asymmetric (Eq. (9))	Symmetric +1/-1
Replay sampling	Class-balanced (50/50)	Uniform
Model selection	Validation, recall-constrained	Last episode
Dropout	0.2	0.3
Teacher forcing	Yes	No
Warm-up pre-fill	25,000 transitions	None
LR schedule	Cosine annealing	Step decay
Batch size	512	256
Training pipeline	Batch-chunk (256)	Single-sample
Loss function	Huber (smooth L1)	Huber (smooth L1)

as it is more interpretable to a deployment engineer than a normalised rate.

C. COMPARISON PROTOCOL

E-DDQN is trained and evaluated directly on the binary Normal-versus-Attack task, whereas the baseline natively predicts the four ordinal grades. To place both on the same footing, the baseline four-class predictions are projected to binary exactly as the labels are (grade 0 $\rightarrow$ Normal; grades 1–3 $\rightarrow$ Attack), so both models are scored against identical binary ground truth on the identical 3,146-sample test set. The baseline native four-class behaviour is reported separately for completeness. Because the two models differ in several training and algorithmic choices at once Table 5, this projection gives a fair head-to-head on identical ground truth but is not a single-variable comparison; component-level attribution is deferred to the ablation in Section V-F.

V. RESULTS AND DISCUSSION

A. TRAINING DYNAMICS

E-DDQN converges rapidly and stably. The per-episode greedy reward rises steeply through roughly the first 100 episodes as the value function exploits the warm-started, class-balanced buffer, then settles onto a stable plateau of about 5,720 reward units for the remainder of the 500-episode run Fig. 8. Test metrics recorded every 10 episodes show recall exceeding 98% within the first 50 episodes and held near 99.5% thereafter, while accuracy, precision, and F1 improve gradually as the policy converts false positives into confirmed detections (Fig. 10); the number of missed attacks falls from the high tens to single digits by mid-training and stabilises at 9–11 (Table 6). This reflects the combined design: the Double DQN target prevents value inflation [13], cosine annealing [38] shrinks the step size near the plateau, and warm-up removes the early instability of single-sample baselines [12].

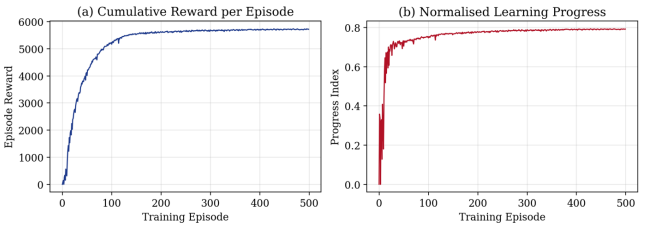


FIGURE 8. E-DDQN training dynamics over 500 episodes: (a) cumulative reward per episode and (b) normalised learning progress.

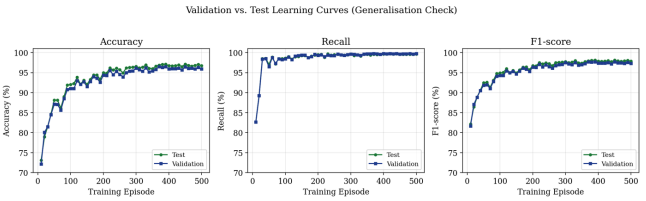


FIGURE 9. Validation versus test learning curves for accuracy, recall, and F1; close agreement across 500 episodes evidences generalization.

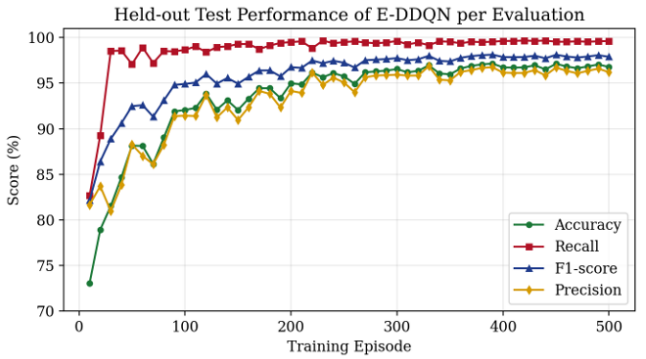


FIGURE 10. Held-out test accuracy, precision, recall, and F1-score across evaluations. Recall rises earliest and highest, reflecting the asymmetric reward.

TABLE 6. E-DDQN test-set performance at representative training checkpoints. Recall rises above 99% early and false negatives fall to single digits by mid-training.

Episode	Acc (%)	Prec (%)	Rec (%)	F1 (%)	TP	TN	FP	FN
50	88.14	88.28	97.07	92.47	2290	483	304	69
100	92.05	91.43	98.64	94.90	2327	569	218	32
150	92.05	90.95	99.28	94.93	2342	554	233	17
200	94.98	94.14	99.49	96.74	2347	641	146	12
250	95.74	95.06	99.49	97.22	2347	665	122	12
300	96.54	95.92	99.62	97.73	2350	687	100	9
350	95.96	95.29	99.53	97.37	2348	671	116	11
400	96.73	96.15	99.62	97.86	2350	693	94	9
450	97.11	96.71	99.53	98.10	2348	707	80	11
500	96.76	96.19	99.62	97.88	2350	694	93	9

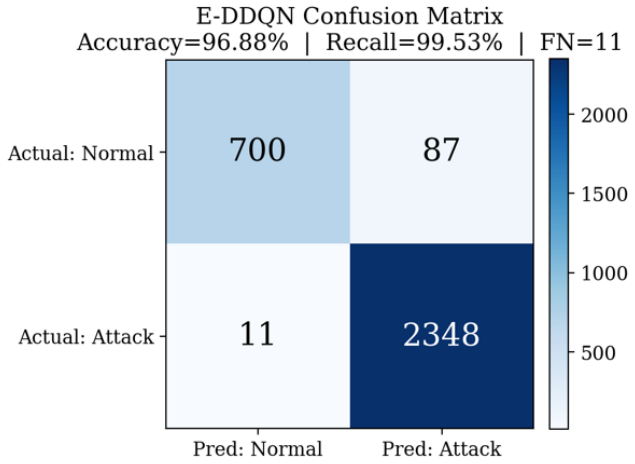


FIGURE 11. Confusion matrix (counts) of E-DDQN on the 3,146-sample test set at the deployed operating point (accuracy 96.88%, recall 99.53%, FN = 11).

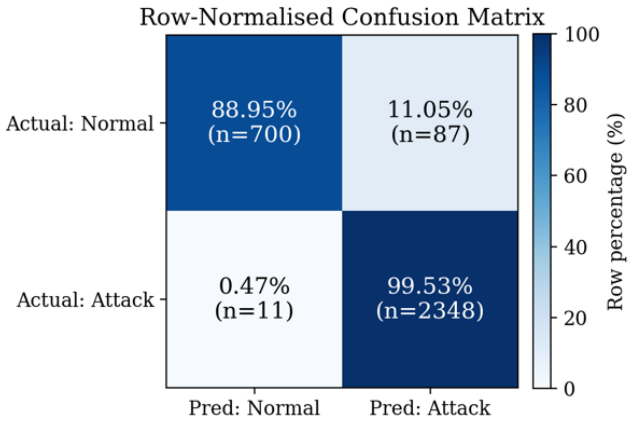


FIGURE 12. Row-normalised confusion matrix of E-DDQN: 99.53% of attacks detected at an 11.05% false-alarm rate on Normal states.

B. E-DDQN DETECTION PERFORMANCE

On the held-out test set of 3,146 samples, the deployed best-validation model attains 96.88% accuracy, 96.43% precision, 99.53% recall, 97.96% F1, and MCC 0.916, with TP = 2,348, TN = 700, FP = 87, FN = 11 (Figs. 11 and 12). It detects 2,348 of 2,359 attacks (11 missed) and clears 700 of 787 benign samples, a high-sensitivity yet well-balanced safety monitor at an 11.05% false-alarm rate. Per-checkpoint values appear in Table 6. Across partitions (Table 7), accuracy is 98.05% (train), 96.47% (validation), and 96.88% (test): a roughly 1.2-point train-to-test gap with closely tracking validation and test curves (Fig. 9) indicates that the high recall generalises rather than reflecting overfitting or majority-class bias.

C. COMPARISON WITH THE STANDARD DQN BASELINE

On the identical binary task the standard DQN attains 94.53% accuracy, 98.07% precision, 94.57% recall, and 96.29% F1 (TP = 2,231, TN = 743, FP = 44, FN = 128). E-DDQN improves accuracy (+2.35), F1 (+1.67), MCC (0.861→0.916), and recall (+4.96) simultaneously, cutting missed attacks from 128 to 11 (−91.4%); the baseline keeps a narrow edge on precision (98.07% vs 96.43%) and specificity (94.41% vs 88.95%) because E-DDQN accepts 87 versus 44 false alarms to reach near-complete recall (Table 8, Fig. 13). Against classical baselines on the same split (Fig. 14), logistic regression and Gaussian naive Bayes reach high recall but collapse on balanced accuracy (61.9% and 59.9%) by over-predicting Attack, whereas E-DDQN leads on balanced accuracy and F1. A class-weighted MLP scores higher in aggregate (98.28% accuracy, 98.85% F1, 0.955 MCC) yet misses 41 attacks (nearly four times E-DDQN's 11), which is the crux of the safety argument: when a missed attack dominates cost, the operating point matters more than mean accuracy. The baseline's native four-class accuracy is 92.12%, but it recalls only 77.2% of grade-1 (mildest) cybersickness; under binary projection these errors become the 128 missed attacks, precisely the borderline region the asymmetric reward targets.

Because E-DDQN and the baseline differ in several respects at once (Table 5), this head-to-head gain should not be attributed to any single mechanism; the leave-one-out ablation (Section V-F, Table 11) supplies that attribution.

D. ROBUSTNESS UNDER CLASS IMBALANCE

With a 75% Attack base rate, aggregate accuracy alone can mislead, so Table 9 reports imbalance-robust metrics. E-DDQN leads on recall (99.53% vs 94.57%), F2 (98.90% vs 95.25%), negative predictive value (98.45% vs 85.30%, so

TABLE 7. Performance of the deployed E-DDQN model across the training, validation, and held-out test partitions (generalization check).

Split	Acc (%)	Prec (%)	Rec (%)	Spec (%)	F1 (%)	MCC	FN
Train	98.05	97.63	99.82	92.72	98.71	0.948	20
Validation	96.47	95.73	99.75	86.64	97.70	0.905	6
Test	96.88	96.43	99.53	88.95	97.96	0.916	11

TABLE 8. Head-to-head comparison on the identical binary test set (3,146 samples). E-DDQN improves accuracy, recall, F1, and MCC simultaneously and reduces missed attacks by 91.4%.

Metric	Standard DQN (binary)	E-DDQN (proposed)	Difference
Accuracy (%)	94.53	96.88	+2.35
Precision (%)	98.07	96.43	−1.64
Recall (%)	94.57	99.53	+4.96
F1 (%)	96.29	97.96	+1.67
MCC	0.861	0.916	+0.055
False negatives	128	11	−117 (−91.4%)
False positives	44	87	+43
Specificity (%)	94.41	88.95	−5.46

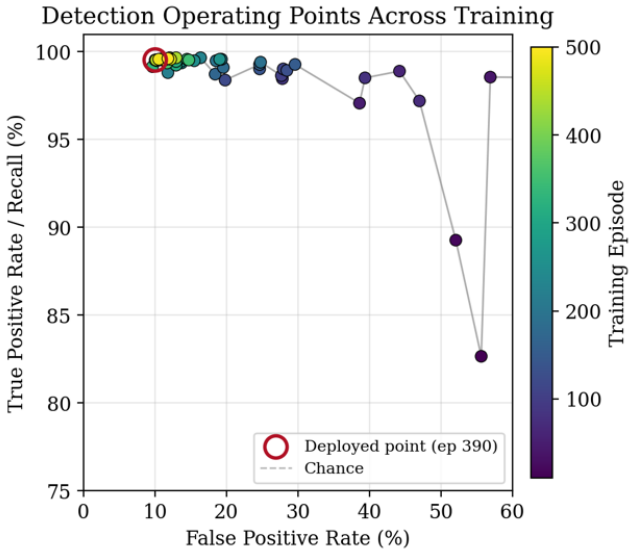


FIGURE 13. Detection operating points across training in recall-false-positive-rate space; the deployed point at episode 390 is circled.

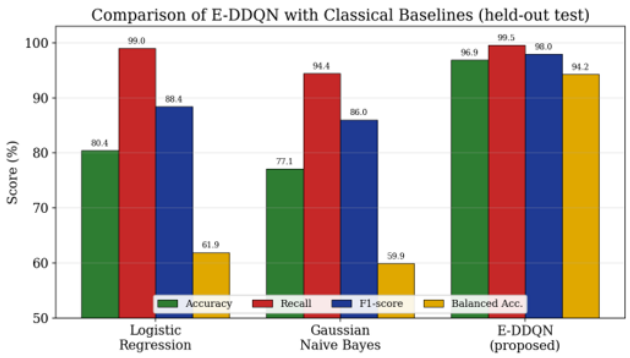


FIGURE 14. Comparison of E-DDQN with classical baselines on the held-out test set: E-DDQN attains the highest accuracy, F1, and balanced accuracy while preserving top recall.

a benign verdict almost never hides an attack), and MCC (0.916 vs 0.861), and is level on balanced accuracy (94.24% vs 94.49%) and G-mean (94.09% vs 94.49%); the baseline retains only a small edge on specificity and precision. Improving the threshold-independent summaries (MCC, F2) while matching balanced accuracy confirms the advantage is genuine rather than a base-rate artefact.

E. STATISTICAL ROBUSTNESS AND SIGNIFICANCE

Retraining the full model from scratch over five seeds (42, 1, 7, 123, 2024) under the identical protocol gives recall 99.62 ± 0.06, accuracy 96.58 ± 0.64, F1 97.76 ± 0.41, MCC 0.908 ± 0.017, and 9 ± 1 missed attacks (Table 10); the deployed single-seed model lies within this distribution. A paired McNemar test against the standard DQN on the shared test set gives $\chi^2 = 28.6$ ($p = 8.8 \times 10^{-8}$), with 96 instances corrected by E-DDQN versus 34 by the baseline, so the improvement is significant at any conventional threshold.

F. ABLATION STUDY

A leave-one-out ablation (seed 42, identical protocol) disables each component in turn and retrains, attributing behaviour to specific mechanisms (Fig. 15, Table 11). The soft target update is critical for stability: hard synchronisation collapses accuracy to 80.96% and specificity to 26.81% (MCC 0.43). Class-balanced replay is the principal driver of aggregate quality: removing it drops accuracy to 94.69% and specificity to 79.03% as the agent reverts to over-alerting. The asymmetric reward is the safety lever: removing it more than triples missed attacks (11→39) and lowers recall to 98.35%, even though aggregate accuracy is marginally higher, exactly the trade-off between accuracy and safety the design intends. The dueling head and warm-up each add small, consistent gains. Teacher forcing is not beneficial on this dataset (re-

TABLE 9. Imbalance-robust evaluation on the binary test set. E-DDQN leads on recall, F2, NPV, F1, and MCC and is level on balanced accuracy and G-mean.

Metric	E-DDQN (proposed)	Standard DQN (binary)
Recall / Sensitivity (%)	99.53	94.57
Specificity (%)	88.95	94.41
Precision / PPV (%)	96.43	98.07
Negative predictive value (%)	98.45	85.30
Balanced accuracy (%)	94.24	94.49
G-mean (%)	94.09	94.49
F1 score (%)	97.96	96.29
F2 score (%)	98.90	95.25
Matthews corr. coef. (MCC)	0.916	0.861

TABLE 10. Mean $\pm$ standard deviation of E-DDQN test metrics over five random seeds, confirming low-variance, reproducible performance.

Metric	Mean $\pm$ SD ($n = 5$ seeds)
Accuracy (%)	96.58 ± 0.64
Precision (%)	95.98 ± 0.80
Recall (%)	99.62 ± 0.06
Specificity (%)	87.47 ± 2.63
Balanced accuracy (%)	93.54 ± 1.30
F1 (%)	97.76 ± 0.41
MCC	0.908 ± 0.017
False negatives	9 ± 1

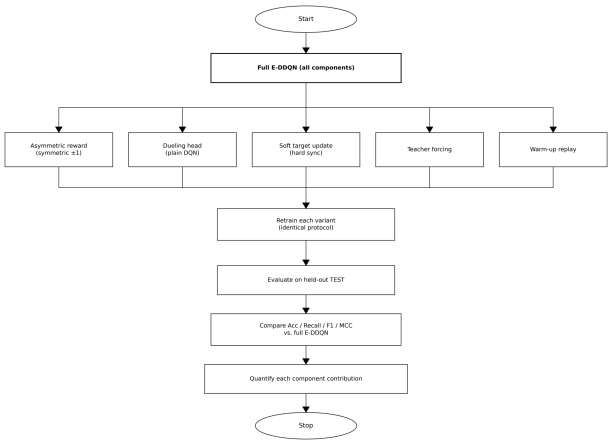


FIGURE 15. Leave-one-out ablation design: each core component is removed in turn and the variant retrained under an identical protocol to quantify its contribution.

moving it leaves performance essentially unchanged), so it is optional and retained only for faster early convergence.

G. THE SAFETY TRADE-OFF

Encoding the unequal cost of the two error types directly in the reward yields an operating point that is simultaneously more accurate and far safer than the symmetric baseline. The cost asymmetry is stark: a false negative lets an adversarial attack reach the user (nausea, disorientation, postural instability, fall risk [1]), whereas a false positive triggers only a brief, recoverable precaution. E-DDQN cuts missed attacks by 91.4% (128→11) while raising accuracy by 2.35 and F1 by 1.67 points, conceding little precision and specificity, the

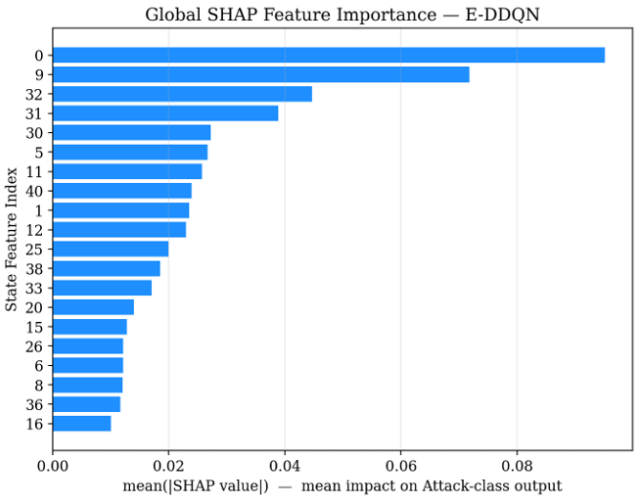


FIGURE 16. Global SHAP feature-importance ranking for E-DDQN: mean absolute contribution of the twenty most influential state dimensions.

same sensitivity-first principle used in other safety-critical alarm design [30], [40]. The 5:1 penalty ratio is tunable: raising it pushes the operating point toward even higher recall, and lowering it recovers baseline-like precision, allowing a deployment to be positioned on the trade-off curve to match its tolerated false-alarm budget.

H. EXPLAINABILITY ANALYSIS

SHAP DeepExplainer shows the learned policy relies on a compact subset of telemetry channels rather than diffuse noise (Fig. 16). The mean absolute Shapley values are dominated by Feature_0 (0.095), followed by Feature_9 (0.072), Feature_32 (0.045), Feature_31 (0.039), and Feature_30 (0.027), with a rapidly decaying tail, a parsimonious, auditable decision rule that the standard DQN cannot provide. Because attribution is post hoc, this transparency adds no inference latency, satisfying the joint explainability-and-real time requirement identified as an open gap in Section II [2], [24].

I. DISCUSSION AND IMPLICATIONS

Three implications follow. First, framing cybersickness detection as a safety-prioritized decision problem under an adversarial threat model is empirically actionable: an RL detector can be trained to suppress missed positive-class events

TABLE 11. Leave-one-out ablation (seed 42). Soft target update and class-balanced replay drive aggregate quality; the asymmetric reward drives the safety-critical reduction in missed attacks.

Variant	Acc (%)	Rec (%)	Spec (%)	F1 (%)	MCC	FN
Full E-DDQN	96.88	99.53	88.95	97.96	0.916	11
− asymmetric reward	97.04	98.35	93.14	98.04	0.921	39
− dueling head	96.44	99.58	87.04	97.67	0.904	10
− soft target update	80.96	99.03	26.81	88.64	0.426	23
− teacher forcing	97.30	99.75	89.96	98.23	0.928	6
− warm-up replay	95.71	99.70	83.74	97.21	0.885	7
− class-balanced replay	94.69	99.92	79.03	96.58	0.858	2

to a handful of cases on realistic VR telemetry. Second, the safety-prioritized objective, not network capacity, is the decisive lever; re-pricing the two error types and adding targeted stability mechanisms cut missed attacks from 128 to 11 while also improving aggregate accuracy and MCC, shifting the design emphasis from ever-larger models toward objective functions that encode domain consequences [30], [40]. Third, post hoc explanation closes the transparency gap within the real time budget, which matters for the regulated clinical and industrial settings where such detectors would be deployed [2]. E-DDQN is thus best read not as a higher-accuracy classifier but as a tunable, auditable safety monitor whose operating point is an explicit, defensible engineering choice. Because the Attack label is a proxy derived from cybersickness-severity grades, these findings establish safety-prioritized detection under an adversarial threat framing rather than demonstrated robustness to explicit input perturbations, which the limitations below and the future-work programme address.

J. LIMITATIONS AND THREATS TO VALIDITY

Several limitations qualify these conclusions. The evaluation uses a single VR telemetry dataset and a fixed split (though now over five seeds with a significance test); cross-dataset and cross-headset studies would further strengthen the claim of generality. A class-weighted supervised baseline attains higher aggregate accuracy than E-DDQN (at the cost of far more missed attacks), so the method is recommended specifically where a missed attack dominates cost. The added false positives lower specificity to 88.95%, whose real cost depends on the user-experience impact of precautionary interventions, which is not measured here. The adversarial labels are proxies derived from discomfort grades, so robustness against explicit FGSM and PGD attacks remains to be demonstrated directly. Finally, SHAP attributions are reported over anonymised feature indices; mapping the dominant features to named physical signals would make the explanations directly actionable for VR engineers. These define a clear programme of future work: multi-dataset validation, an explicit adversarial-attack benchmark, an operating-curve study over the reward ratio, and a user study quantifying the cost of false positives.

VI. CONCLUSION

This study addressed cybersickness detection in immersive VR as a safety-prioritized decision problem motivated by adversarial perceptual threats and proposed E-DDQN, a safety-prioritized reinforcement-learning detector that augments a standard DQN with a dueling Double DQN value head, an asymmetric cost-sensitive reward, class-balanced experience replay, soft target updates, warm-up initialisation, teacher-forced exploration, and validation-based recall-constrained selection, complemented by a post hoc SHAP module. Evaluated against a standard DQN baseline under a common protocol on a 20,972-sample VR telemetry corpus, E-DDQN reduced missed attacks on the held-out test set from 128 to 11 (a 91% reduction) and raised recall from 94.57% to 99.53% while simultaneously improving accuracy (94.53%→96.88%), F1 (96.29%→97.96%), and MCC (0.861→0.916). A roughly 1.2-point train-to-test gap, low variance across five seeds, and a significant McNemar test ($p < 0.001$) confirm the gains are stable and generalise; a leave-one-out ablation isolates the asymmetric reward as the source of the safety gain and class-balanced replay and the soft update as the drivers of aggregate quality. SHAP analysis showed the policy relies on a compact, auditable subset of telemetry channels at no inference-time cost. Future work will extend the evaluation across multiple datasets, add an explicit adversarial-attack benchmark, characterise the full recall-precision operating curve as a function of the reward ratio, and quantify the user-experience cost of false positives, toward a deployable, tunable, and explainable safety monitor for high-risk VR environments. Because the present evaluation uses proxy labels, it constitutes a carefully evaluated step toward dependable safety-aware VR detection rather than a complete perturbation-based adversarial-robustness study, which remains future work.

DATA AVAILABILITY AND REPRODUCIBILITY

The source code for E-DDQN, the standard DQN baseline, and all preprocessing, training, evaluation, and figure-generation scripts will be made available by the corresponding author upon reasonable request, together with the exact data partitions and the trained model weights. The GLCD dataset was obtained from its original providers and is available from them on request. All experiments fix the random seed (42) and enable deterministic CUDA settings, so every

reported figure and table can be reproduced exactly.

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CONFLICT OF INTEREST

The authors declare no conflict of interest.

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